

Drauflassen

From German brewing and more

Drauflassen (German for letting something flow onto) is a common practice in large and small breweries. Especially when the fermenter capacity greatly exceeds the brew house capacity or when fresh yeast is introduced into the brewery. It allows pitching at a proper pitching rate with a reduced amount of yeast. Cooled and aerated wort is added to fill only $\frac{1}{3}$ – $\frac{1}{2}$ of the fermenter capacity. Then the yeast is pitched. After 24 hrs, when first signs of fermentation are visible and the first batch is at low Kraeusen the 2nd batch, which has been fully aerated, is added. Hence this technique is known as Double Batch in American brewing. Depending on how small the initial batch was, this process can be repeated a number of times. Narziss reports that this technique is beneficial to the attenuation and the ester levels of the final beer. The latter are reduced because the yeast is kept longer in its growth phase during which it consumes the ester precursor acetyl CoA [Narziss, 2005]

This technique can also be used by home brewers to deal with an insufficient yeast amount to pitch a full batch of wort. While commercial brewers brew the 2nd batch shortly before it is needed, this approach is less practical for the home brewer as this would mean 2 back-to-back brew days while each brew day would only produce half the amount of wort needed. A better approach is sanitary storage of part of the wort. This is less problematic for lagers than for ales as the fermentation temperature and thus the storage temperature is lower. But lagers are the styles of beers that are more likely in need of this technique anyway as they require a higher pitching rate.

Here is how it can be done and how I have done it in the past:

Cool the wort with extra attention to sanitation. That for example means that the pot is kept covered as soon as the boil stops. Use a whirlpool to settle out the trub and some of the cold break. Now transfer about $\frac{1}{3}$ to $\frac{1}{2}$ of the wort to the fermentation vessel and aerate it in there. I don't like to aerate the complete volume at this point in order to minimize oxidation processes that could take place while the rest is sitting unpitched. If you use a bucket or carboy aeration through shaking is an attractive option b/c of the large headspace and the lower weight. Then the yeast is pitched and a sample for the Fast Ferment Test is taken.

The fermenter is placed into my fermentation fridge and the remainder of the wort stays in the pot. That gets closed with a lid and a trash bag is used to cover it up. The latter is to prevent contaminants to be blown in through the holes I have in the lid for the immersion chiller lines. I intentionally don't transfer into another vessel as that would require more cleaning and sanitizing and just increases the risk of infection. This pot is then placed into the fermentation fridge as well where it will sit at the same temperature as the beer.

After 24hrs I expect to see low Kraeusen on the beer. Now it is time to add the remainder of the wort. Aeration may be done in the kettle or once it has been added to the fermenter. Aeration in the kettle will disturb the trub and you will have to let it settle again while aeration in the fermenter may release large amounts of CO₂ that can cause the beer to foam over. More recently I have started to aerate in the fermenter and have not experienced major foam-ups. Once the transfer is complete, the fermenter goes back into the fermentation fridge and after a few days it should be at high Kraeusen.

In order to assess the potential of getting an infection by having stored unpitched wort for an extended time you may want to do a sanitation or forced ferment test. In this test you sanitize a small vessel by boiling or other means and add a sample of the wort that has been sitting unpitched to it. Keep this sample covered and at a warm place. The time it takes to show signs of spontaneous fermentation (i.e. gets cloudy) will tell how sanitary the wort still was. Anything that remains clear for greater than 2-3 days was sanitary enough.

If the beer is not showing signs of fermentation after 24 hours you should still add the rest of the wort. At this point there is an increased risk that the unpitched wort will spoil. You may add some dry yeast at this point and ensure better yeast viability for the next batch. If you use a counterflow or plate chiller you would need to transfer the wort into 2 sanitized vessels: one for pitching and fermentation and the other for temporary storage.

While some eyebrows might be raised about the idea of introducing oxygen to already fermenting beer (the 2nd batch is fully aerated) I doubt that the general oxidation concerns exist here as aeration of the additional batches of wort is practice in the industry and such concerns would have been mentioned.

Another interesting practice found in some commercial breweries is a form of constant Drauflassen [Kunze 2007]:

A batch of wort is pitched. Once it is at high Kraeusen part of it is pumped to another vat and fresh aerated wort is added to replace the young beer that was drawn off. The yeast will quickly absorb the newly added oxygen and continue to grow. Once it is at high Kraeusen again the process is repeated.

This method allows for a constant pitching with very young, clean and fresh yeast that is kept in the growth phase in the "Kraeusen vat". At the end of the week the "Kraeusen vat" is cleaned and then pitched again at the beginning of the next week.

While this method has a lot of benefits for the fermentation and the health of the pitched yeast it is impractical for the home brewer since home brewers don't brew frequent enough and also don't brew one single recipe over and over.

References

[Narziss, 2005] Prof. Dr. agr. Ludwig Narziss, Prof. Dr.-Ing. habil. Werner Back, Technische Universitaet Muenchen (Fakultaet fuer Brauwesen, Weihenstephan), *Abriss der Bierbrauerei*. WILEY-VCH Verlags GmbH Weinheim Germany, 2005

[Kunze, 2007] Wolfgang Kunze, *Technologie Brauer und Maelzer*, 9. Auflage, VLB Berlin

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